Assignment 7

October 31, 2025

8-4 Problem Statement

Consider the gas-phase parallel reactions operated at 27 °C and 4 atm with pure A in the feed (volumetric flow ($= 10\,\text{dm}^3, \text{min}^{-1}$)):

A $\xrightarrow{k_1}$ X, $\quad r_X = k_1 C_A^{1/2}, \quad k_1 = 0.004; (\text{mol}, \text{dm}^{-3})^{1/2}, \text{min}^{-1}$ A $\xrightarrow{k_2}$ B,$r_B \ = k_2 C_A$,$k_2 \ = 0.3; \text{min}^{-1}$ A $\xrightarrow{k_3}$

B is desired; X and Y are pollutants.

(a) Instantaneous Selectivities vs ($\mathbf{C}_A$)

$$S_{B/X} = \frac{r_B}{r_X} = \frac{k_2 C_A}{k_1 C_A^{1/2}} = \frac{k_2}{k_1} C_A^{1/2}, \ S_{B/Y} \ = \frac{r_B}{r_Y} = \frac{k_2}{k_3} \frac{1}{C_A}, \ S_{B/(X+Y)} = \frac{r_B}{r_X + r_Y} = \frac{k_2 C_A}{k_1 C_A^{1/2} + k_3 C_A^2}. \tag{1}$$

Maximizing ($\text{S}_{B/(X+Y)})withrespectto(C_A) : let(x! =!C_A^{1/2})sothat(S = k_2 x/(k_1+k_3 x^3)).Then(\frac{dS}{dx} \propto k_1 - 2k_3 x^3 = 0 \Rightarrow x^3 = k_1/(2k_3)).Hence \boxed{C_A^* = \left(\frac{k_1}{2k_3}\right)^{2/3}} \quad \Rightarrow \quad C_A^* = 0.04; \text{mol}, \text{dm}^{-3}.(2)$
This value is used as the target concentration in the first reactor.

(b)–(d) First Reactor in Series (CSTR at ($\mathbf{C}_A = C_A^*$))

Feed concentration at (27 °C,;4 atm): with ideal gas ($\text{C}_{A0} = P/RT), (C_{A0} = 4/(0.082057 \cdot 300.15) = 0.162; \text{mol}, \text{dm}^{-3}).For a steady CSTR with pure A feed, V_1 = \dot{V}, \frac{C_{A0} - C_A}{r_X + r_B + r_Y}\Big|_{C_A = C_A^*}$.(3)
At ($\text{C}_A^* = 0.04) : (r_X = 0.0008), (r_B = 0.012), (r_Y = 0.0004) mol, dm^{-3}$,$\text{min}^{-1}$, so

$$\boxed{V_1 = 9.27 \times 10^1; \text{dm}^3}, \quad \boxed{\tau_1 = V_1/\dot{V} = 9.27; \text{min}}. \tag{4}$$

Effluent concentrations (from species balances ($\text{C}_i = \tau_1 r_i), no inlet of X, B, Y) : C_X = 7.42 \times 10^{-3}, ; C_B = 1.11 \times 10^{-1}, ; C_Y = 3.71 \times 10^{-3}; (\text{mol}, \text{dm}^{-3})$.(5) Conversion of A: ($\text{X}_1 = (C_{A0} - C_A^*)/C_{A0} = 0.754$; (75.4

(f) Temperature Recommendation for Single CSTR

Given $E_1 = 20\,000\,\text{cal},\text{mol}^{-1}$, $E_2 = 10\,000\,\text{cal},\text{mol}^{-1}$, $E_3 = 30\,000\,\text{cal},\text{mol}^{-1}$, $\tau = 10\,\text{min}$, and $C_{A0} = 0.1\,\text{mol},\text{dm}^{-3}$, rate constants vary by Arrhenius about the $27\,°\text{C}$ base values. Because $E_3 > E_1 > E_2$, increasing T accelerates pollutant formation more than the desired reaction; therefore the recommended T is the lowest that meets performance constraints. The MATLAB scan returns the T that maximizes $S_{B/(X+Y)}$ for this CSTR and reports the steady-state concentrations.

(g) Pressure Selection (1–100 atm)

At fixed T, $C_{A0} \propto P$. Here $S_{B/X} \propto C_A^{1/2}$ increases with P, while $S_{B/Y} \propto 1/C_A$ decreases with P. The best pressure for a single CSTR is the one that yields an internal C_A near C_A^*. MATLAB sweeps $P \in [1, 100]$ atm to display $S_{B/(X+Y)}$ at the CSTR steady state.

8-10. Problem Description

Note:- All the calculations and graphical analysis is available in the MATLAB CODE. Terephthalic acid (TPA) and its potassium salt are key intermediates in the production of synthetic fibers such as Dacron and Mylar. The formation of potassium terephthalate (S) from potassium benzoate (A) proceeds via intermediates (R) through both series and autocatalytic reactions. The reaction mechanism is:

$$A \xrightarrow{k_1} R \quad R \xrightarrow{k_2} S \quad R + S \xrightarrow{k_3} 2S$$

Here:

- A = Potassium benzoate (K-benzoate)
- R = Lumped intermediates (K-phthalates, K-isophthalates, etc.)
- S = Potassium terephthalate (K-terephthalate)

2. Kinetic Data

The specific reaction constants at 410°C and activation energies are provided as follows:

Parameter	k_i (at 410°C)	Units	E_i (kcal/mol)
k_1	1.08×10^{-3}	s^{-1}	42.6
k_2	1.19×10^{-3}	s^{-1}	48.6
k_3	1.59×10^{-3}	$\text{dm}^3\ \text{mol}^{-1}\ \text{s}^{-1}$	32.0

The temperature dependence of each rate constant follows the Arrhenius equation:

$$k_i(T) = k_{i,\text{ref}} \exp\left[-\frac{E_i}{R}\left(\frac{1}{T} - \frac{1}{T_{\text{ref}}}\right)\right], \quad R = 0.001987 \text{ kcal mol}^{-1}\text{K}^{-1} \tag{6}$$

3. Batch Reactor Model

For a constant-volume, isothermal batch reactor:

$$\frac{dC_A}{dt} = -k_1 C_A \quad \frac{dC_R}{dt} = k_1 C_A - k_2 C_R - k_3 C_R C_S \quad \frac{dC_S}{dt} = k_2 C_R + k_3 C_R C_S \tag{7}$$

with initial conditions $C_A(0) = C_{A0} = 1$, $C_R(0) = 0$, and $C_S(0) = 0$.

These equations were numerically integrated in MATLAB using `ode45`. Simulations were conducted at 390°C, 410°C, and 430°C.

Results

The concentration profiles of A, R, and S show a clear intermediate accumulation. The time and value at which R reaches its maximum were determined numerically.

Temperature (°C)	k_1	k_2	k_3	$t_{\max R}$ (s)	$R_{\max}$
390	4.19×10^{-4}	4.04×10^{-4}	7.81×10^{-4}	1842	0.335
410	1.08×10^{-3}	1.19×10^{-3}	1.59×10^{-3}	728	0.321
430	2.64×10^{-3}	3.30×10^{-3}	3.11×10^{-3}	288	0.307

As expected, higher temperature accelerates the reaction and shifts the R maximum to earlier times.

4. CSTR Model

At steady state for a CSTR with pure A feed ($C_{A0} = 1$ mol/dm^3) and space time $\tau = 1200$ s:

Material Balances

$$0 = \frac{C_{A0} - C_A}{\tau} - k_1 C_A \quad 0 = -\frac{C_R}{\tau} + k_1 C_A - k_2 C_R - k_3 C_R C_S \quad 0 = -\frac{C_S}{\tau} + k_2 C_R + k_3 C_R C_S \tag{8}$$

From the A balance:

$$C_A = \frac{C_{A0}}{1 + k_1 \tau} \tag{9}$$

From the S balance:

$$C_S = \frac{\tau k_2 C_R}{1 - \tau k_3 C_R}, \quad (1 - \tau k_3 C_R > 0) \tag{10}$$

Substituting into the R balance yields a quadratic in C_R:

$$k_3 C_R^2 - \left(\frac{1}{\tau} + k_2 + k_1 C_A \tau k_3\right) C_R + k_1 C_A = 0 \tag{11}$$

Steady-State Solution at 410°C

Solving this quadratic gives the physically admissible root:

$$C_A = 0.43554 \ C_R \qquad\qquad = 0.17836 \ C_S = 0.38610$$

These values were computed using the MATLAB script $\texttt{p8}_10b_cstr.m$.

5. Conclusions

- The intermediate R forms and decays faster with increasing temperature.
- The autocatalytic term $k_3C_RC_S$ accelerates the final formation of S once sufficient S is present.
- In the CSTR, C_A is reduced to about 0.44, with substantial steady-state formation of S.
- The MATLAB models accurately capture the kinetics and allow direct comparison of reactor behavior under varying conditions.